

Track 2G Curve-Aware Training Campaign Results

Track 2 Campaign Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `Track 2G` curve-aware training campaign. The campaign goal was to test whether adding curve-aware loss terms to the harmonic residual-offset architecture improves the scalar training surface before returning the candidates to official Track 2 curve-first verification.

The campaign completed all planned entries

- `12` completed runs;
- `0` failed runs;
- `4` loss profiles;
- `3` required direction surfaces: `global`, `Fw`, and `Bw`.

The runner-level scalar first entry is `te_track2g_curve_aware_raw_centered_shape_fw`, but this is not a single deployment winner. The closeout keeps one best candidate for each required surface. Official Track 2 verification remains a separate operator-approved step.

Campaign Artifacts

Artifact	Path
Campaign output	<code>output/training_campaigns/2026-06-08-18-36-30_track2g_curve_aware_training_campaign_2026_06_08</code>
Leaderboard	<code>output/training_campaigns/2026-06-08-18-36-30_track2g_curve_aware_training_campaign_2026_06_08/campaign_leaderboard.yaml</code>
Best-run pointer	<code>output/training_campaigns/2026-06-08-18-36-30_track2g_curve_aware_training_campaign_2026_06_08/campaign_best_run.yaml</code>
Execution report	<code>output/training_campaigns/2026-06-08-18-36-30_track2g_curve_aware_training_campaign_2026_06_08/campaign_execution_report.md</code>
Planning report	<code>doc/reports/campaign_plans/track2/2026-06-08-18-01-40_track2g_curve_aware_training_campaign_plan_report.md</code>
Campaign package	<code>config/training/track2g_curve_aware_training/campaigns/2026-06-08_track2g_curve_aware_training_campaign</code>

Execution Summary

Field	Value
Campaign name	track2g_curve_aware_training_campaign_2026_06_08
Completed runs	12
Failed runs	0
Model type	curve_aware_harmonic_residual_offset_probe
Loss profiles	pointwise_control , raw_centered_shape , raw_offset , full_curve_composite
Tested surfaces	global , Fw , Bw
Runner-level scalar first entry	te_track2g_curve_aware_raw_centered_shape_fw
Program-level scalar winner changed	no

Directional Branch Results

Surface	Candidate	Loss Profile	Test MAE	Test RMSE	Val MAE
global	full_curve_composite_global	full_curve_composite	0.003345	0.003713	0.003616
Fw	raw_centered_shape_fw	raw_centered_shape	0.003181	0.003571	0.003251
Bw	pointwise_control_bw	pointwise_control	0.003430	0.003945	0.003749

These three rows are the branch-level closeout result. The Fw row is also the campaign scalar leader, but it does not replace the global or Bw branches.

Loss Profile Leaderboard

Rank	Surface	Candidate	Loss Profile	Test MAE	Test RMSE	Val MAE
1	Fw	raw_centered_shape_fw	raw_centered_shape	0.003181	0.003571	0.003251
2	Fw	full_curve_composite_fw	full_curve_composite	0.003260	0.003630	0.003320
3	Fw	raw_offset_fw	raw_offset	0.003279	0.003698	0.003328
4	global	full_curve_composite_global	full_curve_composite	0.003345	0.003713	0.003616
5	global	raw_centered_shape_global	raw_centered_shape	0.003350	0.003753	0.003636
6	Fw	pointwise_control_fw	pointwise_control	0.003371	0.003763	0.003291
7	Bw	pointwise_control_bw	pointwise_control	0.003430	0.003945	0.003749
8	global	raw_offset_global	raw_offset	0.003465	0.003829	0.003564
9	Bw	raw_centered_shape_bw	raw_centered_shape	0.003465	0.003998	0.003740
10	Bw	raw_offset_bw	raw_offset	0.003471	0.003992	0.003751
11	Bw	full_curve_composite_bw	full_curve_composite	0.003511	0.004113	0.003803
12	global	pointwise_control_global	pointwise_control	0.003587	0.004001	0.003607

Baseline Comparison

Surface	Track 2G Best	Track 2G MAE	Track 2F-Bis Ref.	Ref. MAE	Delta
global	full_curve_composite	0.003345	T2F-bis clean global	0.003528	+5.20%
Fw	raw_centered_shape	0.003181	T2F-bis harmonic Fw	0.002862	-11.14%
Bw	pointwise_control	0.003430	T2F-bis harmonic Bw	0.003336	-2.83%

Scalar training metrics show a mixed result. The full composite loss improves the global Track 2G scalar branch versus the previous Track 2F-bis global control. The Fw and Bw branches do not beat the strongest Track 2F-bis direction-specific harmonic candidates on scalar test_mae .

This does not yet decide curve-first promotion. Track 2G was designed to test curve-aware behavior, so the next evidence must come from official Track 2 curve playback, overlays, offset diagnostics, and mean-centered shape checks.

Loss Interpretation

Loss Profile	Observed Signal	Interpretation
pointwise_control	Best Bw Track 2G scalar result.	The curve-aware add-ons are not automatically beneficial on the backward branch.
raw_centered_shape	Best Fw Track 2G scalar result and campaign scalar winner.	Centered-shape pressure is useful for forward scalar training, but not enough to beat Track 2F-bis harmonic Fw .
raw_offset	No branch winner.	Offset-only pressure did not dominate the first scalar campaign.
full_curve_composite	Best global Track 2G scalar result.	Combining pointwise, centered-shape, offset, amplitude, and sparse harmonic terms is most promising for the global branch.

The main modeling result is not "Track 2G wins". The useful result is narrower: the composite objective is promising for global , centered-shape is promising for Fw , and Bw still resists the first curve-aware loss profiles.

Registry Effects

The campaign runner refreshed family registries for all twelve Track 2G families and updated the program registry. The program-level scalar best did not move: `te_periodic_gru_sequence_remote_Bw` remains the current scalar program winner with test `MAE = 0.002344`.

Registry Scope	Best Run	Test MAE
Track 2G global composite	<code>full_curve_composite_global</code>	0.003345
Track 2G Fw centered-shape	<code>raw_centered_shape_fw</code>	0.003181
Track 2G Bw pointwise	<code>pointwise_control_bw</code>	0.003430

Track 2 Boundary

Official Track 2 curve-first verification was not run as part of this normal campaign closeout. Under campaign governance, that remains a separate operator-approved workflow after this campaign-results report and PDF are complete.

The next Track 2 package should add all twelve Track 2G candidates and report the accepted result separately for:

- `global` ;
- `Fw` ;
- `Bw` .

The verification must compare raw curve error, centered-shape error, offset, amplitude, harmonic behavior, collage plots, and overlay plots against Track 2F, Track 2F-bis, Wave 2B, and the current accepted Track 2 baselines.

Closeout Decision

Track 2G is complete from an execution standpoint: all planned candidates have successful training artifacts, no failed run remains, registries were refreshed by the runner, and the active campaign state can be cleared.

From a modeling standpoint

- carry `full_curve_composite_global` forward as the strongest Track 2G global scalar candidate;
- carry `raw_centered_shape_fw` forward as the strongest Track 2G forward scalar candidate;
- carry `pointwise_control_bw` forward as the strongest Track 2G backward scalar candidate;
- do not promote Track 2G over Track 2F-bis or Wave 2B before official Track 2 verification.

Recommended Follow-Up

1. Accept this closeout and clear the active campaign state.
2. Prepare a separate operator-launched Track 2 verification refresh for all twelve Track 2G candidates.
3. Use the Track 2 result to decide whether Track 2G should continue as a loss-only branch or whether the next campaign should move to the explicit multi-head shape/offset architecture.
4. Keep `Fw` , `Bw` , and `global` as parallel branch decisions.